

Executive Summary

This proposal outlines an approach by which our organisation can use customer data, machine learning and industry best practices to materially improve marketing efforts by increasing the success rates of outbound calls to reducing campaign costs and improve commercial returns.

Our Business Challenge

Currently, our organisation undertakes a traditional approach to outbound marketing and sales. Call lists are generated from our Customer Relationship Management (CRM) system using manually defined criteria and passed to our outbound call centre for execution.

Success with this approach is on par with industry outcomes (in the order of 2-6%)¹ where success is defined as customer acceptance of offer and commencement of the application process.

The cost of this approach for 10,000 customer contacts, yielding 500 successful sales can be calculated using the following metrics in AUD:

Call Type	Call Count	Call Durations	Cost (\$50 p/h)	Revenue
Unsuccessful	9500	5 Minutes	\$39,583	0
Successful	500	15 Minutes	\$6,250	\$1,550,000.00
			\$45,833	Cost per customer \$91.67

A modest improvement in call success rates from 5% to 10% of calls, would offer significant value generation, achieving a target of 1000 successful calls much earlier than undertaking 10,000 total calls would permit campaigns to conclude significantly earlier, permitting retasking or additional outbound campaigns.

Call Type	Call Count	Call Durations	Cost (\$50 p/h)	Revenue
Unsuccessful	9000 (Full campaign may not be necessary)	5 Minutes	\$37,500	0
Successful	1000	15 Minutes	\$12,500	\$3,100,000.00
			\$50,000	Cost per customer \$50.00

¹ <https://towardsdatascience.com/why-every-marketer-should-consider-uplift-modeling-1090235572ec>

Why machine learning, what are our goals?

Machine learning permits us to learn from experience by analysing a body of data to derive rules to predict future data², the McKinsey Global Institute³ research indicates that the field of banking AI can improve performance by 50% over other analytics techniques, with a net industry benefit of improving revenues between 2.5 and 5.2% industry wide.

Business Goals

- Drive an improvement in a positive response rate for outbound calls resulting in shorter campaign duration through faster achievement of uptake targets.
- Continually improve based on daily experience.
- Establish a technical capability which can be applied other offers, products, and communication channels
- Be ethically sound and compliant.

Within the following constraints of

- Existing data held by the organisation about individuals.
- In observance of the Australian Privacy Principles⁴ and to the National Credit Code⁵

² <https://workofthefuture.mit.edu/wp-content/uploads/2020/12/2020-Research-Brief-Malone-Rus-Laubacher2.pdf>

³

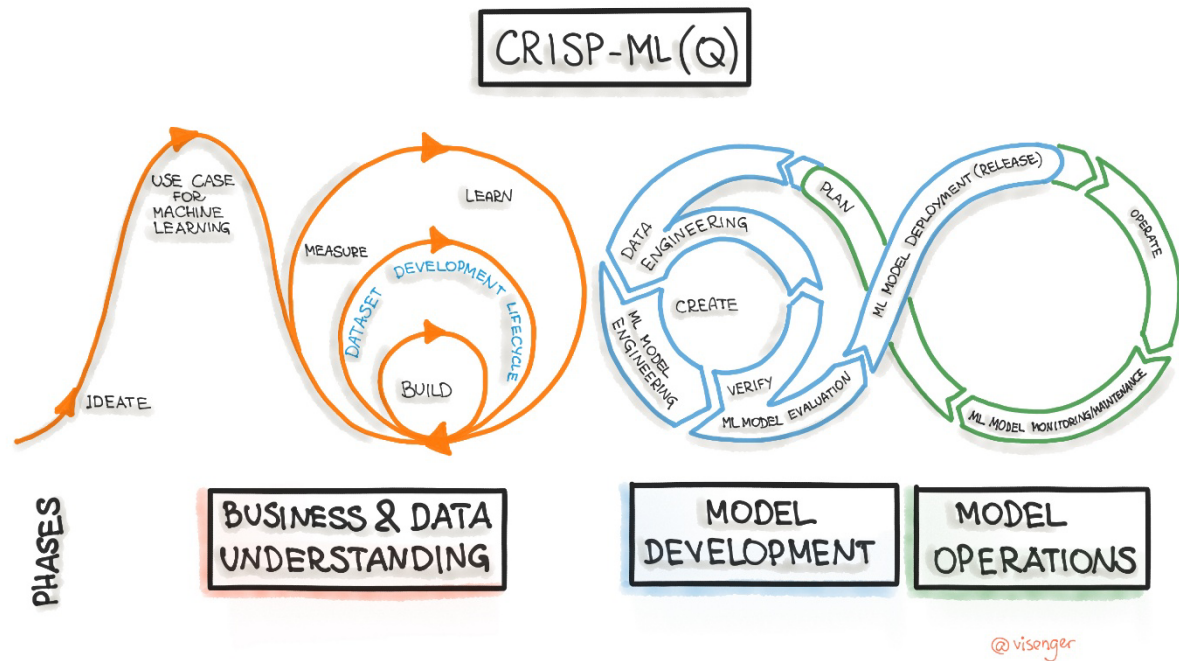
<https://www.mckinsey.com/~media/mckinsey/featured%20insights/artificial%20intelligence/notes%20from%20the%20ai%20frontier%20applications%20and%20value%20of%20deep%20learning/notes-from-the-ai-frontier-insights-from-hundreds-of-use-cases-discussion-paper.pdf>

⁴ <https://www.oaic.gov.au/privacy/australian-privacy-principles>

⁵ <https://asic.gov.au/regulatory-resources/credit/credit-general-conduct-obligations/national-credit-code/>

Introducing CRISP-ML, an industry standard model for Machine Learning Quality

CRISP-ML is a proposed methodology consisting of the following framework for the adoption of a Machine Learning development. We will use this framework to outline our proposed process.



Business and Data Understanding

We have summarised the overall concepts using the Machine Learning Canvas, “Plan on a Page” approach to describe key concepts, the canvas is outlined below

PREDICTION TASK Type of task? Entity on which predictions are made? Possible outcomes? Wait time before observation? For an uncontacted set of customers generate a probability score indicating likelihood of offer acceptance.	DECISIONS How are predictions turned into proposed value for the end-user? Use the prediction score as a ranking mechanism to prioritise call order for the following day. Call customers in the prioritised order, recording call outcomes.	VALUE PROPOSITION Who is the end-user? What are their objectives? How will they benefit from the ML system? Mention workflow/interfaces. Call Centre Agents, seeking a higher rate of success when making outreach calls during working hours of 11am to 7pm Workflow: Customer Contact lists will be generated by the ML system and created as Leads within CRM. Leads will be distributed to agents using existing services and use a versioned call script to run the call	DATA COLLECTION Strategy for initial train set & continuous update. Mention collection rate? Extraction of customer data from existing CRM and core banking system with summary. Previous days call outcomes to be provided to the system to provide retraining, include call script versions.	DATA SOURCES Where can we get (raw) information on entities and observed outcomes? Mention database tables, API methods, websites to scrape, etc. Core banking/CRM repositories of Customer information and product holdings extracted at start of campaign Daily call outcomes from CRM activities.
OFFLINE SIMULATION Simulation of the impact of decision/predictions? Which test data? Cost/gain values? Deployment criteria (min performance value, fairness) Test data = Outcomes of manual call execution indicating a single week of calling against a randomized set of candidate customers. Updated with previous days call outcomes. Candidate model supplants production model if predictions outperforms > 5%	MAKING PREDICTIONS When do we make real-time / batch pred.? Time available for this + featurization + post-processing? Compute target? Predictions are made nightly post call centre close via automated extraction and execution of ML Ops Pipeline. Call candidates prepared by start of business following day.	Agents will record call outcome and used script version during call wrap, duration will be recorded by the CTI system. Assuming ML program success Campaign goals (target sign ups) should complete some days earlier than traditional means.	BUILDING MODELS How many prod models are needed? When would we update? Time available for this? 1 Production Model updated nightly with current day actual call outcomes. - n-Candidate models updated and maintained in parallel, with production deployment after evaluation. 16 hours for automated evaluation and morning manual review.	FEATURES Input representations available at prediction time, extracted from raw data sources. - Gender - Contact Script - Job - Marital Status - Age - Postcode - Education - Home Ownership - Call Duration - Products (Loan)
	MONITORING Metrics to quantify value creation and measure the ML system's impact in production (on end-users and business)? Successful Duration of successful calls			

Data Engineering

Target candidates for this solution will be existing direct banking customers (as opposed to institutional, small business and other segments) who do not hold credit card products (either direct cards or cards associate with mortgage holdings).

Candidates will be filtered based on completeness and washed to remove or eliminate personal data (for instance, names and addresses by be replaced with a customer identifier and a postcode or Australian Statistical Geography Standard (ASG) coding). Candidate attributes will be combined product holding data (For instance, mortgage holding, or customer life-time value).

Where features are largely incomplete or outdated (for instance, last known credit score) for the data set, they will be excluded from use in the target models

Finally, with the use of target models, a wrapper method such as backward selection can be used to eliminate unneeded features, eliminating the need for any supporting data preparation from final models.

ML Model Engineering

We propose a logistic regression, yielding a probabilistic continuous output between (0.0 and 1.0) The logistic model will emit a continuous value for each prediction, permitting calls scheduled in a descending priority based on capacity of the outbound call system in each day.

Discrete categorisation ('call/no call') was considered, using logistic regression or a decision trees. But would generate an erroneous consultation that our ML model is an absolute predictor of human behaviour rather than a probability indicator.

Training will occur using an input data from manually executed campaigns. Initial data for manual campaign selected for appropriate statistical value (mixture gender, age, location and other demographics) and contain actual outcomes.

The evaluation steps (outlined below) will be undertaken repeatedly during the experimentation phase and automated daily during production operation in line with our ML Ops strategy.

ML Model Evaluation and Deployment

Given the batch nature of our solution, it's use in a marketing context, and the relatively low data volumes (10,000) the evaluation metrics should focus primarily on accuracy rather than explainability, latency or robustness. We propose the initial use of Mean Absolute Error as an evaluation metric for individual predictions and the use of a per day measure of expected revenue based upon average probabilities for a day's prediction.

Within each model, tuning of hyper parameters will be assisted with cross-validation, using the k-fold cross validation procedure and use of the training data. Given the relatively low volume of data we are making use of in this environment, a value of k for 5, resulting in a data set of ~200 values per fold however this will be adjusted based on candidate data size.

Once experimentation has yielded several candidate models, the highest performing model will become our Production model, with up to 3 other deployment candidates being retained as

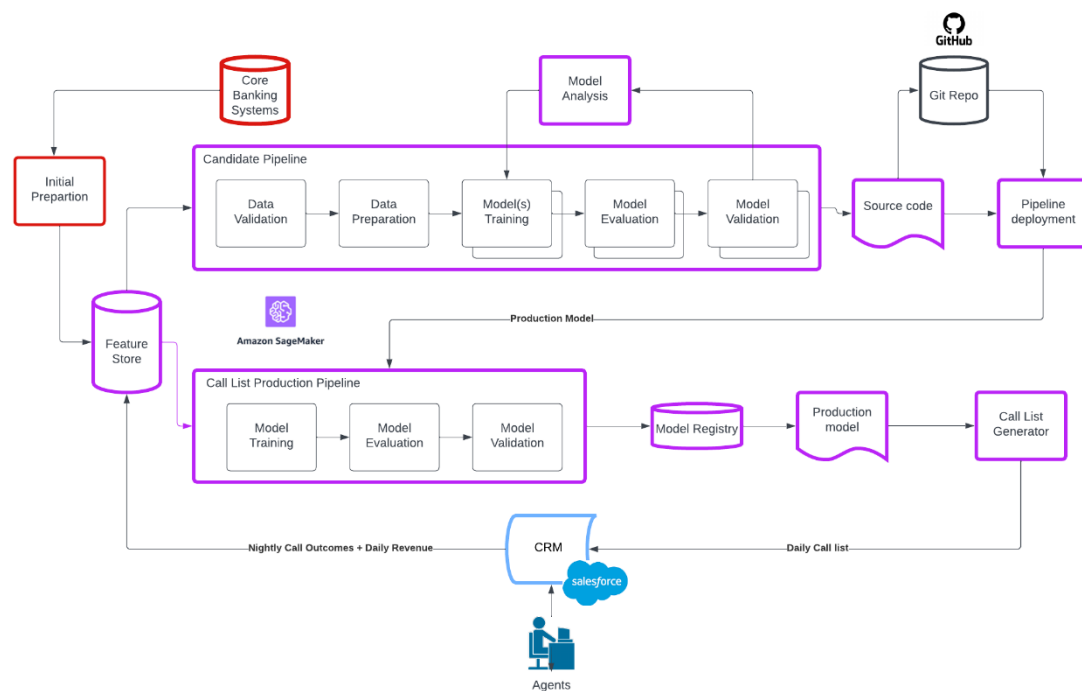
Candidate model's for continued evaluation within an automated pipeline, refer to MLOp's below.

ML Model Monitoring

Once within the MLOps pipeline model monitoring will occur daily via automated means, assessing model accuracy for predictions. In addition, after several days of operation, daily revenue actuals will be known, permitting the creation of a regression model against which production and candidate models can be assessed for daily revenue performance.

ML Ops

We propose an MLOps pipeline design with significant automation. Initial data is extracted from Core banking systems and preparation is undertaken at the commencement of a campaign. Once target customer lists have been enriched, they are passed to a feature store and enter an automated Candidate Pipeline where multiple candidate models are evaluated. The successful Production model is deployed to the Call List Production pipeline for training with production data and deployment. Daily call lists are generated, and outcomes are passed back for continuous training and back to the candidate pipeline process. Where appropriate production models are replaced with better performing candidates based on outcomes from Model Monitoring.



Recommended Tech Stack

Our organisation has a significant technical investment with Amazon Web Services running secure workloads in the Sydney region, as such we propose the use of the Amazon SageMaker stack. Amazon SageMaker is a managed machine learning service designed to simplify the process of building, training, and deploying machine learning models and will eliminate the need for us to consider complex infrastructure.

A mapping of product to DevOps and MLOps capabilities is outlined below.

Practice	DevOps	MLOps	Proposed Tech Stack
Version Control	Code version Control	Code version control + Data versioning + Model versioning (linked for reproducibility)	Github + Amazon Sagemaker Model Registry
Pipeline	n/a	Training ML Pipeline Serving ML Pipeline	Amazon Sagemaker Pipelines
Behaviour Validation	Unit tests	Model validation	Amazon Sagemaker
CI/CD	Deploys code to production	Deploys code to production + training ML pipeline	Amazon Sagemaker Pipelines
Data validation	n/a	Statistical validation	Amazon SageMaker Data Wrangler
Monitoring	SLO-Based	SLO + differential monitoring	Amazon SageMaker Model Monitor